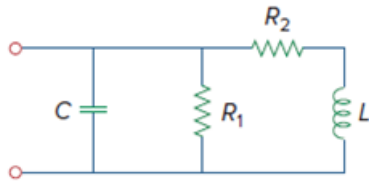


Problem

Using Fig. 14.80, design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of RLC circuits.

Figure 14.80:



Step-by-step solution

Step 1 of 1

Using Fig. 14.80, design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of an RLC circuit.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in Example 14.9.

Problem

For the circuits in Fig. 14.80, find the resonant frequency ω_0 , the quality factor Q , and the bandwidth B . Let $C = 0.1$ F, $R_1 = 10$ Ω , $R_2 = 2$ Ω , and $L = 2$ H.

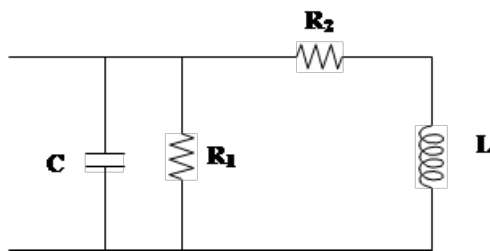


Figure 14.80

For Prob. 14.41.

Solution

To find ω_0 , we need to find the input impedance or input admittance and set imaginary component equal to zero. Finding the input admittance seems to be the easiest approach.

$$Y = j\omega 0.1 + 0.1 + 1/(2 + j\omega 2) = j\omega 0.1 + 0.1 + [2/(4 + 4\omega^2)] - [j\omega 2/(4 + 4\omega^2)]$$

At resonance,

$$0.1\omega = 2\omega/(4 + 4\omega^2) \text{ or } 4\omega^2 + 4 = 20 \text{ or } \omega^2 = 4 \text{ or } \omega_0 = \mathbf{2 \text{ rad/s}}$$

and,

$$Y = 0.1 + 2/(4 + 16) = 0.1 + 0.1 = 0.2 \text{ S}$$

The bandwidth is defined as the two values of ω such that $|Y| = 1.4142(0.2) = 0.28284$ S.

I do not know about you, but I sure would not want to solve this analytically. So how about using MATLAB or excel to solve for the two values of ω ?

Using Excel, we get $\omega_1 = 1.414$ rad/s and $\omega_2 = 3.741$ rad/s or $B = \mathbf{2.327 \text{ rad/s}}$

We can now use the relationship between ω_0 and the bandwidth.

$$Q = \omega_0/B = 2/2.327 = \mathbf{0.8595}$$